

MCQ's

(lower limb)

Anterior & medial compartments of thigh, gluteal region

M.C.Q'S

- 1- Regarding the fascia of the thigh, mark the correct statements:**
 - a- The fascia lata of thigh is thickened medially to form iliotibial tract.
 - b- The saphenous opening is a gap in the fascia lata.
 - c- The saphenous opening is 6 cm below and lateral to pubic tubercle.
 - d- Two facial septa pass from fascia lata to the femur.
- 2- Regarding the iliotibial tract:**
 - a- It is attached between iliac crest and lateral condyle of the tibia.
 - b- It receives the insertion of gluteus medius and tensor fascia lata.
 - c- It belongs to the investing layer of the deep fascia of the thigh.
 - d- It is connected to linea aspra by the medial intermuscular septum.
- 3- The iliotibial tract receives attachment of the following muscle:**
 - a- Tensor fasciae lata muscle.
 - b- Gluteus medius muscle.
 - c- Iliacus.
 - d- Gluteus minimus.
- 4- The cribriform fascia is pierced by:**
 - a- Great saphenous vein.
 - b- Superior epigastric artery.
 - c- Deep circumflex iliac artery.
 - d- Obturator.
- 5- One of following structures pass deep to the inguinal ligament:**
 - a- Spermatic cord.
 - b- Psoas major muscle.
 - c- Ilioinguinal nerve.
 - d- Great saphenous vein.
- 6- Which of the following muscles is a flexor of the thigh:**
 - a- Superior gemellus.
 - b- Adductor longus.
 - c- Gracilis.
 - d- Psoas major.
- 7- The sartorius muscle:**
 - a- Flexes the knee joint and extends the hip joint.
 - b- Is one of the boundaries of the femoral triangle.
 - c- Arises from the anterior inferior iliac spine.
 - d- Is fusiform muscle.

- 8- **The sartorius muscle is :**
- a- It is innervated by the obturator nerve.
 - b- It arises from the anterior superior iliac spin.
 - c- It inserts on the lateral surface of the tibia.
 - d- It forms the medial border of the femoral triangle.
- 9- **A crushing blow that breaks the anterior superior iliac spine would damage the origin of the:**
- a- Biceps femoris muscle.
 - b- Pectineus muscle.
 - c- Rectus femoris muscle.
 - d- Sartorius muscle.
- 10- **The head of the quadriceps muscle which acts on the hip joint:**
- a- Vastus lateralis muscle.
 - b- Rectus femoris.
 - c- Vastus intermedius.
 - d- Vastus medialis.
- 11- **Muscle that extends leg include the following muscle:**
- a- Biceps femoris.
 - b- Rectus femoris.
 - c- Semitendinosus.
 - d- Gracilis.
- 12- **Regarding quadriceps femoris muscle, choose the correct answer:**
- a- Extends the knee joint.
 - b- Flexes the knee joint.
 - c- Extends the hip joint.
 - d- Abducts the hip joint.
- 13- **The articularis genu muscle, mark the correct statement:**
- a- Is the deep fibers of the vastus medialis muscle.
 - b- Is fixed to the upper border of the suprapatellar bursa.
 - c- Is supplied by the obturator nerve.
 - d- Arises from the upper half of the medial surface of the femur.
- 14- **One of the following muscles is supplied by the obturator nerve:**
- a- Psoas major.
 - b- Sartorius.
 - c- Iliacus.
 - d- Adductor brevis.

15- Femoral vessels enter popliteal fossa through hiatus in which muscle:

- a- Adductor longus.
- b- Rectus femoris.
- c- Adductor magnus.
- d- Gracilis.

16- The boundaries of the femoral triangle is by the following:

- a- Adductor longus, pectineus and sartorius.
- b- Inguinal ligament, sartorius and adductor longus.
- c- Inguinal ligament, sartorius and adductor brevis.
- d- Inguinal ligament, sartorius and adductor magnus.

17- The base of the femoral triangle is formed by the:

- a- Sartorius muscle.
- b- Adductor longus muscle.
- c- Inguinal ligament.
- d- Pubic tubercle.

18- In floor of femoral triangle one of the following muscles is seen:

- a- Adductor magnus.
- b- Iliacus.
- c- Vastus medialis.
- d- Tensor fascialata.

19- The floor of the femoral triangle is bounded medially by:

- a- Pectineus.
- b- Adductor longus.
- c- Iliacus.
- d- Psoas major.

20- In the femoral triangle one of the following branches arise from the femoral artery:

- a- Superior epigastric.
- b- Obturator artery.
- c- Superficial circumflex iliac.
- d- Deep circumflex iliac

21- The femoral sheath:

- a- Its medial compartment is occupied by the femoral artery.
- b- Its lateral compartment is occupied by the femoral nerve.
- c- The intermediate compartment is occupied by the femoral vein.
- d- The anterior wall is formed by fascia iliaca.

22- Concerning the femoral sheath, choose the correct answer:

- a-** It encloses the proximal portions of the popliteal artery and vein.
- b-** Posteriorly it is formed by the fascia transversalis.
- c-** The femoral nerve is medial to the femoral sheath.
- d-** The inguinal ligament is anterior to the femoral sheath.

23- Femoral canal contains:

- a-** Femoral artery.
- b-** Femoral vein.
- c-** Femoral nerve.
- d-** Lymphatic vessels.

24- The femoral ring is bounded medially by:

- a-** Lacunar ligament.
- b-** Inguinal ligament.
- c-** Pubic ramus.
- d-** Femoral artery.

25- Following form the medial boundary of femoral ring:

- a-** Femoral septum.
- b-** Lacunar ligament.
- c-** Fascia transversalis.
- d-** Cribriform fascia.

26- The femoral ring is bounded anteriorly by:

- a-** Femoral vein.
- b-** Lacunar ligament.
- c-** Obturator artery.
- d-** Inguinal ligament.

27- A femoral hernia descends through the femoral canal, and the neck of the hernial sac lies:

- a-** At the saphenous opening.
- b-** Above and medial to the pubic tubercle.
- c-** At the femoral ring.
- d-** In the obturator canal.

28- Concerning the femoral hernia, choose the correct answer:

- a-** It is more common in males than females.
- b-** It occurs in the medial compartment of the femoral sheath.
- c-** It is found above and medial to the pubic tubercle.
- d-** Its neck is medial to the base of the lacunar ligament.

29- Regarding the femoral sheath and femoral hernia, the following statement is true:

- a- Femoral sheath is a prolongation of fascial lining from the abdominal walls, along the upper part of femoral vessels.
- b- Femoral canal is the lateral compartment of the femoral sheath.
- c- Femoral hernia is more common in men than in women.
- d- Neck of the femoral hernia lies below and medial to the pubic tubercle.

30- The roof of the adductor canal is formed by one of the followings:

- a- Rectus femoris.
- b- Adductor longus.
- c- Vastus medialis.
- d- Sartorius.

31- One of the following structures is in the adductor canal:

- a- Femoral artery.
- b- Femoral lymph node.
- c- Profunda femoris artery.
- d- Great saphenous vein.

32- The following structures pass through the subsartorial canal from upper to lower end is:

- a- Posterior division of the obturator nerve.
- b- Saphenous nerve.
- c- Femoral artery.
- d- Nerve to the vastus intermedius.

33- Regarding the adductor (subsartorial) canal, choose the correct answer:

- a- Vastus lateralis muscle in its lateral aspect.
- b- Nerve plexus embedded in its posterior wall.
- c- Descending branch from femoral artery which takes part in the anastomosis around the knee.
- d- Nerve to the vastus lateralis muscle.

34- Regarding the adductor (subsartorial) canal and its contents, mark one correct statement:

- a- Its anteromedial wall is formed by the vastus medialis.
- b- Its lateral wall is formed by the vastus lateralis.
- c- Its posterior wall is formed by the adductor longus and magnus.
- d- The femoral artery has no branches in the canal.

35- Contents of the adductor (subsartorial) canal include one of the following:

- a- Great saphenous vein.
- b- Saphenous nerve.
- c- Superficial inguinal lymph nodes.
- d- Obturator artery.

36- Regarding adductor canal, the following statement is correct:

- a- It ends at the adductor hiatus.
- b- It transmits the femoral nerve.
- c- Its posterior wall is formed by the adductor brevis muscle only.
- d- Its lateral wall is formed by the vastus lateralis.

37- The femoral nerve originates from which of the following roots?

- a- T12, L1, L2.
- b- L1, L2, L3.
- c- L2, L3, L4.
- d- L3, L4, L5.

38- One muscle is supplied by the femoral nerve in the abdomen:

- a- Iliacus.
- b- Vastus lateralis.
- c- Sartorius.
- d- Vastus medialis.

39- Damage to the femoral nerve on the lateral wall of the pelvis would affect the function of the following muscle:

- a- Sartorius.
- b- Gracilis.
- c- Adductor longus.
- d- Adductor brevis.

40- The femoral nerve:

- a- Comes from the dorsal divisions of the L2, 3 & 4.
- b- Ends four inches below the inguinal ligament.
- c- Supply all muscles of floor of femoral triangle except the pectineus.
- d- Passes to thigh through lateral compartment of the femoral sheath.

41- Concerning the femoral nerve it:

- a- Enters the thigh medial to the femoral sheath.
- b- Supplies motor nerves to most of the muscles of the anterior and medial compartments of the thigh.
- c- Lies medial to the femoral artery in the femoral triangle.
- d- Has no motor branches below the knee.

42- Which of the following nerves innervates at least one muscle that acts on both the hip and the knee joints:

- a- Ilioinguinal nerve.
- b- Femoral nerve.
- c- Saphenous nerve.
- d- Common peroneal nerve.

43- Loss of extension of the knee would result from damage to the:

- a- Femoral nerve.
- b- Tibial nerve.
- c- Tibial and common peroneal nerves.
- d- Obturator and femoral nerves.

44- Paralysis of saphenous nerve results in the following deformity:

- a- Inability to invert the foot.
- b- Loss of sensation on the lateral malleolus.
- c- Loss of sensation on the medial malleolus.
- d- Loss of sensation on localized area on the dorsum of the foot.

45- Regarding the patellar tendon reflex (knee jerk), mark one correct statement:

- a- It is the flexion of the knee joint on tapping the patellar tendon.
- b- It is a test for second, third and fourth lumbar spinal nerves.
- c- It is a test for the hamstrings muscles.
- d- It is a test for the calf muscles.

46- The obturator nerve:

- a- Supplies the hip joint.
- b- Supplies the knee joint.
- c- Gives a cutaneous branch to the lower medial side of the thigh.
- d- Arises from lumbar plexus.

47- The obturator nerve supplies the following muscle:

- a- Sartorius.
- b- Iliacus.
- c- Gracilis.
- d- Rectus femoris.

48- Regarding the obturator nerve, mark correct answer:

- a- Is formed by ventral divisions of 2, 3 and 4 lumbar nerves.
- b- Supplies ankle joint.
- c- Crosses the internal iliac artery.
- d- Leaves the pelvis through femoral sheath.

49- The obturator nerve; mark the most suitable answer:

- a- Is formed of the ventral divisions of L2, 3 & 4 nerves.
- b- Is completely motor.
- c- Supplies all the adductor and flexor thigh muscles.
- d- It passes through the adductor canal.

50- The tensor fascia lata, muscle:

- a- Attached to the posterior part of iliac crest.
- b- Inserted into the iliotibial tract.
- c- Innervated by the inferior gluteal nerve.
- d- Lies deep to the gluteus medius muscle.

51- The most powerful extensor of the hip joint is:

- a- Semimembranosus.
- b- Gluteus maximus.
- c- Sacrospinalis.
- d- Gluteus medius.

52- The powerful extension of the thigh, required when one is standing from a sitting position, is a function of the:

- a- Gluteus maximus muscle.
- b- Psoas major muscle.
- c- Iliacus muscle.
- d- Obturator externus muscle.

53- Following muscle produce abduction of hip joint:

- a- Gluteus medius.
- b- Obturator internus.
- c- Superior gemellus.
- d- Quadratus femoris.

54- Abductors of the hip joint are:

- a- Called into action in sitting.
- b- Their paralysis cause waddling gait.
- c- Gluteus medius and maximus.
- d- Supplied by inferior gluteal nerve.

55- Medial rotation of the thigh is done the following muscle:

- a- Gluteus maximus.
- b- Quadratus femoris.
- c- Gluteus minimus.
- d- Superior gemellus.

- 56- A significant weakness in thigh abduction involve which muscle:**
- a- Gluteus medius.
 - b- Pectineus.
 - c- Gracilis.
 - d- Adductor magnus.
- 57- In walking, hip bone of the suspended leg is raised by which of the following muscles acting on the supported side of the body:**
- a- Gluteus maximus.
 - b- Obturator internus.
 - c- Gluteus medius.
 - d- Obturator externus.
- 58- Paralysis of the gluteus medius and minimus will result in the:**
- a- Inability to adduct the thigh at the hip joint.
 - b- Sinking of pelvis on the normal side when same foot is raised.
 - c- Inability to laterally rotate the thigh at the hip joint.
 - d- Waddling gate in case of bilateral paralysis of the muscles.
- 59- Regarding action of the glutei muscles, one answer is correct:**
- a- Gluteus maximus is medial rotator of the thigh.
 - b- Gluteus medius is adductor of the hip joint.
 - c- Both glutei maximus and medius are extensors of the hip joint.
 - d- Both glutei medius and minimus are powerful abductor of hip joint.
- 60- One of the following muscles is lateral rotator of the thigh:**
- a- The iliopsoas.
 - b- The rectus femoris.
 - c- The biceps femoris.
 - d- The obturator internus.
- 61- The quadratus femoris is inserted into, choose the correct answer:**
- a- Greater trochanter of the femur.
 - b- Lesser trochanter of the femur.
 - c- Middle of trochanteric line.
 - d- Quadrate tubercle.
- 62- The following structure deep lies to the gluteus maximus:**
- a- Iliofemoral ligament.
 - b- Obturator nerve.
 - c- Straight head of rectus femoris muscle.
 - d- Greater trochanter of femur.

63- The structure which passes through greater sciatic foramen is:

- a- Quadratus femoris.
- b- Sciatic nerve.
- c- Tendon of obturator externus muscle.
- d- Nerve to obturator internus.

64- Which structure passes through the lesser sciatic foramen:

- a- Piriformis muscle.
- b- Inferior gluteal nerve.
- c- Tendon of obturator internus muscle.
- d- Sciatic nerve.

65- The following structure pass through the lesser sciatic foramen:

- a- Superior gluteal artery.
- b- Sciatic nerve.
- c- Obturator internus tendon.
- d- Pudendal nerve.

1- B	29- A	57- C
2- C	30- D	58- D
3- A	31- A	59- D
4- A	32- C	60- D
5- B	33- C	61- D
6- D	34- C	62- D
7- B	35- B	63- B & D
8- B	36- A	64- C
9- D	37- C	65- C & D
10- B	38- A	
11- B	39- A	
12- A	40- A	
13- B	41- D	
14- D	42- B	
15- C	43- A	
16- B	44- C	
17- C	45- B	
18- B	46- all	
19- B	47- C	
20- C	48- A	
21- C	49- A	
22- D	50- B	
23- D	51- B	
24- A	52- A	
25- B	53- A	
26- D	54- B	
27- C	55- C	
28- B	56- A	

FILL IN THE BLANKS

- 1- Among the structures passes through the saphenous opening are and
- 2- The iliotibial tract is a lateral thickening of theand attached above to the of the iliac crest.
- 3- Muscles attached to iliotibial tract are and
- 4- Sartorius arises from spine while, the straight head of rectus femoris arises from spine.
- 5- Rectus femoris can the hip joint while it can the knee joint.
- 6- Adductor longus is inserted into and it is supplied by nerve.
- 7- The muscle lies between the anterior and posterior divisions of the obturator nerve.
- 8- Adductor magnus is supplied by and nerves.
- 9- At the apex of the femoral triangle the muscle crosses the muscle.
- 10- The anterior wall of the femoral sheath is formed by while the posterior one is formed by
- 11- The femoral canal is to the femoral vein.
- 12- The femoral canal is bounded anteriorly by and medially by
- 13- The posterior wall of the adductor canal is formed above by muscle and below by
- 14- The upper end of the adductor canal transmits the following nerves and
- 15- nerve emerges from the anterior surface of the psoas major muscle while nerve emerges from its medial margin.
- 16- Nerves appear at the medial border of the psoas major muscle and
- 17- The nerve supply the quadriceps muscle.
- 18- The cutaneous nerve which supplies the medial side of leg and foot is it is a branch of
- 19- The obturator nerve supplies the group of muscles.
- 20- The nerve arises from the anterior divisions of the anterior primary rami of L2, 3, 4 is while the nerve arises from the posterior divisions of the same nerve roots.
- 21- The superficial three fourths of the gluteus maximus is inserted into the..... while the deep fourth is inserted into the
- 22- Gluteus maximus is supplied by nerve while gluteus minimus is supplied by nerve.
- 23- Abductors of the hip joint are and
- 24- Superior gemellus is supplied by nerve while the inferior gemellus is supplied by nerve
- 25- The obturator externus muscle is inserted into the
- 26- The superior gluteal nerve ends by supplying the muscle.

- 27- The root value of the sciatic nerve is while that of the inferior gluteal nerve is
 - 28- The lateral cutaneous nerve of thigh is a branch of while posterior cutaneous nerve of thigh is a branch of
 - 29- 2 cutaneous branches from the sacral plexus and
 - 30- 2 nerves pass from the greater to the lesser sciatic foramina are and
 - 31- 2 bony prominences lie deep to gluteus maximus: and
 - 32- 2 ligaments lie deep to gluteus maximus and
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- 1- Great saphenous vein, superficial epigastric artery.
- 2- Fascia lata, iliac tubercle.
- 3- Gluteus maximus, tensor fascia lata.
- 4- Anterior superior iliac, anterior inferior iliac.
- 5- Flex, extend.
- 6- Linea aspera, obturator nerve.
- 7- Adductor brevis.
- 8- Obturator and Sciatic nerves.
- 9- Sartorius, adductor longus.
- 10- Fascia transversalis, fascia iliaca.
- 11- Medial.
- 12- Inguinal ligament, lacunar ligament.
- 13- Adductor longus, adductor magnus.
- 14- Saphenous nerve and nerve to vastus medialis.
- 15- Genitofemoral – obturator.
- 16- Obturator nerve and lumbosacral trunk.
- 17- Femoral.
- 18- Saphenous – femoral.
- 19- Adductor.
- 20- Obturator – femoral.
- 21- Iliotibial tract, gluteal tuberosity.
- 22- Inferior gluteal – superior gluteal.
- 23- Gluteus medius and minimus.
- 24- Nerve to obturator internus – nerve to quadratus femoris.
- 25- Trochanteric fossa.
- 26- Tensor fascia lata.
- 27- L4,5, S1,2,3 – L5, S1,2.
- 28- Lumbar plexus – sacral plexus
- 29- Posterior cutaneous nerve of thigh – perforating cutaneous nerve.
- 30- Pudendal nerve-nerve to obturator internus.
- 31- Greater trochanter – ischial tuberosity.
- 32- Sacrotuberous – sacrospinous.

CROSS MATCHING

Select appropriate statement of column (B) which fits with knowledge of column (A):

1-	Column (A)	Column (B)
	a- Vastus medialis relation. () b- Rectus femoris action. () c- Semitendinosus action. () d- Gracilis nerve supply. () e- Vastus medialis nerve supply. ()	1- A boundary of adductor canal. 2- A flexor of the knee. 3- A flexor of the hip. 4- Supplied by obturator nerve. 5- Femoral nerve.

2-	Column (A)	Column (B)
	a- Femoral nerve. () b- Superior gluteal nerve. () c- Lateral popliteal nerve. () d- Inferior gluteal nerve. ()	1- Gluteus maximus. 2- Short head of biceps femoris. 3- Tensor fascia lata. 4- Pectineus.

3-	Column (A)	Column (B)
	a-Tendon passes through lesser sciatic foramen only. () b-Nerve passes through greater sciatic foramen only. () c-Nerve passes through both sciatic foramina. () d-Nerve passes away of these foramina. ()	1-Internal pudendal nerve. 2- Tendon of obturator internus muscle. 3- Superior gluteal nerve. 4- Obturator nerve.

Match column A with B:

4-	(A)	(B)
	1- Iliacus 2- Sartorius 3- Vastus lateralis 4- Adductor longus 5- Rectus femoris	a- Has two heads b- Originates from linea aspera c- Flexes the knee d- Flexes the hip e- Supplied by obturator nerve

Match the root values of the following nerves:

5-	(A)	(B)
	1- Femoral nerve 2- Sciatic nerve 3- Genito-femoral nerve 4- Nerve to piriformis 5- Posterior cutaneous nerve of thigh	a- L2, 3, 4 b- S1, 2 c- L4, 5 & S1, 2, 3 d- S1, 2, 3 e- L1, 2

1-	a- 1	b- 3	c- 2	d- 4	e- 5
2-	a- 4	b- 3	c- 2	d- 1	
3-	a- 2	b- 3	c- 1	d- 4	
4-	1- d	2- c	3- b	4- e	5- a
5-	1- a	2- c	3- e	4- b	5- d

PROBLEM SOLVING

-1-

A 55 year-old woman presented with a globular swelling in her left groin. She stated that the swelling became smaller when she lay down but nerve completely disappeared.

1. What is your possible diagnosis?
2. What is meant by femoral ring and femoral canal?
3. Why the condition is common in females rather than males?
4. What are the common complications of such condition?

1. Femoral hernia.
2. Femoral canal is a potential space in the medial compartment of femoral sheath while femoral ring is the opening of femoral canal to the abdomen.
3. Due to the wider femoral canal and ring due to greater breadth of female pelvis with relative small femoral vessels.
4. Strangulation of the hernia is a common complication.

**Posterior compartment of thigh, popliteal fossa,
posterior & anterior compartments of leg**

M.C.Q'S

- 1- Regarding the hamstring muscles (Posterior compartment of thigh), mark correct statement:**
 - a- All are supplied by common peroneal nerve.
 - b- They can extend the knee.
 - c- Biceps femoris produces medial rotation of knee.
 - d- Upper half of semimembranosus is membranous.
- 2- Extension of the hip joint is done by the following muscle:**
 - a- Iliacus muscle.
 - b- Sartorius muscle.
 - c- Semimembranosus muscle.
 - d- Rectus femoris muscle.
- 3- Lateral intermuscular septum of the thigh separates which of the following muscles:**
 - a- Adductor longus and gracilis.
 - b- Semitendinosus and gracilis.
 - c- Psoas major and pectineus.
 - d- Vastus lateralis and biceps femoris.
- 4- The sciatic nerve supplies one of the following muscles:**
 - a- Semimembranosus.
 - b- Vastus medialis.
 - c- Rectus femoris.
 - d- Quadratus femoris.
- 5- To avoid sciatic nerve injury, an intramuscular injection in the gluteal region is given in which of the following quadrants?**
 - a- Upper outer quadrant.
 - b- Lower outer quadrant.
 - c- Upper inner quadrant.
 - d- Lower inner quadrant.
- 6- The structure contribute to the above and lateral boundary of the popliteal fossa:**
 - a- Semimembranosus muscle.
 - b- Plantaris.
 - c- Biceps femoris.
 - d- Medial head of the gastrocnemius muscle.

- 7- Regarding popliteal fossa, the following is correct statement:**
- a- Upper part of floor of fossa is formed of popliteal surface of the tibia.
 - b- It contains the medial cutaneous nerve of the thigh.
 - c- Popliteal artery pulsations can be felt while the knee is extended.
 - d- Semitendinosus forms its superior medial boundary.
- 8- The following structure attaches to the calcaneus:**
- a- Tibialis posterior tendon.
 - b- Gastrocnemius tendon.
 - c- Flexor hallucis longus tendon.
 - d- Popliteus muscle.
- 9- The superficial muscles of the posterior compartment of the leg insert on which of the following bones:**
- a- Talus.
 - b- Navicular.
 - c- Calcaneus.
 - d- Tibia.
- 10- The superficial calf muscles; select the correct statement:**
- a- Plantar flex the ankle joint.
 - b- Invert the foot.
 - c- Enters the foot by passing deep to the flexor retinaculum.
 - d- Are five in number.
- 11- Regarding the gastrocnemius muscle, mark the correct answer:**
- a- Its medial head is attached to the popliteal surface of the femur above its medial condyle, it is the smaller head.
 - b- Its lateral head is attached to the femur above the medial epicondyle.
 - c- It is a plantar flexor at the ankle joint.
 - d- It receives its nerve supply from the anterior tibial nerve.
- 12- The muscle which act as plantar flexors of the foot include:**
- a. Extensor hallucis longus.
 - b. Extensor digitorum longus.
 - c. Tibialis anterior.
 - d. Soleus.
- 13- Regarding muscles of posterior compartment of leg, select the correct:**
- a- They are all supplied by anterior tibial nerve.
 - b- They receive their blood supply by anterior tibial artery.
 - c- They produce plantar flexion of foot.
 - d- Tibialis posterior can evert foot.

14- Concerning the flexor hallucis longus muscle, choose the correct answer:

- a- Grooves the lower surface of the calcaneus.
- b- An important factor in maintaining the transverse arch of the foot.
- c- Lies superficial to the flexor retinaculum.
- d- Is supplied by posterior tibial nerve.

15- Unlocking of the knee joint to permit flexion is caused by the action of which muscle:

- a- Vastus medialis.
- b- Articularis genu.
- c- Biceps femoris.
- d- Popliteus.

16- Which of the following is an intra-articular tendon?

- a- Sartorius.
- b- Semitendinosus.
- c- Anconeus.
- d- Popliteus.

17- Regarding tibialis posterior muscle, indicate the correct statement:

- a- It arises from tibia, fibula and interosseous membrane.
- b- Its tendon passes superficial to that of flexor digitorum longus.
- c- It produces inversion of the foot by acting at the ankle joint.
- d- It helps in suspending the lateral longitudinal arch of the foot.

18- Regarding the posterior compartment of the leg, the following statement is correct:

- a- Posterior tibial nerve crosses the posterior tibial artery from lateral to medial.
- b- Pulsation from the posterior tibial artery is felt behind the lateral malleolus.
- c- Plantaris can be used as autograft.
- d- The peroneal artery is the smallest of the branches of posterior tibial artery.

19- Nerve commonly injured in fractured neck of the fibula is the:

- a- Sciatic nerve.
- b- Superficial peroneal nerve.
- c- Deep peroneal nerve.
- d- Common peroneal nerve.

20- Effects of common peroneal nerve injury include one:

- a- Foot drop and inversion of the foot.
- b- Foot drop and eversion of the foot.
- c- Dorsi flexion and eversion of the foot.
- d- Dorsi flexion and inversion of the foot.

21- Damage to posterior tibial nerve would affect following muscle:

- a- Flexor digitorum longus.
- b- Abductor hallucis.
- c- Flexor digitorum brevis.
- d- Flexor hallucis brevis.

22- Following structure is located in anterior compartment of leg:

- a- Posterior tibial artery.
- b- Soleus.
- c- Extensor digitorum longus.
- d- Tibial nerve.

23- Which of the following muscles dorsiflexes foot at ankle joint:

- a- Peroneus longus.
- b- Extensor digitorum brevis.
- c- Tibialis posterior.
- d- Tibialis anterior.

24- The following muscle evert the foot:

- a- Tibialis anterior.
- b- Extensor hallucis longus.
- c- Flexor hallucis longus.
- d- Peroneus tertius.

25- As the deep peroneal nerve descends through the leg, it is joined by which of the following arteries:

- a- Sural.
- b- Posterior tibial.
- c- Middle genicular.
- d- Anterior tibial.

26- One of the following muscles is supplied by the deep peroneal nerve:

- a- Tibialis anterior.
- b- Tibialis posterior.
- c- Flexor digitorum longus.
- d- Peroneus brevis.

27- Following structure passes deep to the extensor retinaculum:

- a-** Tibialis anterior.
- b-** Flexor digitorum longus.
- c-** Posterior tibial vessels.
- d-** Posterior tibial nerve.

1-	D
2-	C
3-	D
4-	A
5-	A
6-	C
7-	D
8-	B
9-	C
10-	A
11-	C
12-	D
13-	C
14-	D
15-	D
16-	D
17-	A
18-	C
19-	D
20-	A
21-	A
22-	C
23-	D
24-	D
25-	D
26-	A
27-	A

FILL IN THE BLANKS

- 1- The biceps femoris muscle has long head that arises from and short head arises from
 - 2- The biceps femoris is inserted into the and acts as a of the knee joint.
 - 3- Biceps femoris is inserted into while semimembranosus is inserted into
 - 4- The sciatic nerve leaves the pelvis by passing through below the muscle.
 - 5- The sciatic nerve is divided into and nerves below the middle of the back of thigh.
 - 6- Popliteal fossa is bounded above and medially by 2 muscles and
 - 7- The popliteal fossa is bounded above and laterally by muscle while it is bounded below and laterally by muscle.
 - 8- The nerve can be palpated against the neck of the fibula.
 - 9- Sural nerve is a branch of nerve while sural communicating nerve is a branch of
 - 10- The common peroneal nerve has 2 cutaneous branches and
 - 11- The motor effect of the common peroneal nerve injury are: and
 - 12- The is inserted into the posterior surface of the calcaneus.
 - 13- The popliteus muscle does of the knee joint at the beginning of
 - 14- Tibialis anterior muscle is inserted into and
 - 15- Among the anterior group of muscles of the leg muscle causes inversion while muscle causes eversion of the foot.
-

- 1- Ischial tuberosity, lateral lip of linea aspera.
- 2- Head of fibula, flexor.

- 3- Head of fibula – groove on the back of medial condyle of tibia.
- 4- Greater sciatic foramen, piriformis.
- 5- Tibial and common peroneal nerves.
- 6- Semitendinosus – semimembranosus.
- 7- Biceps femoris – lateral head of gastrocnemius.
- 8- Common peroneal.
- 9- Tibial – common peroneal nerve.
- 10- Sural communicating nerve – lateral cutaneous nerve of calf.
- 11- Foot drop – loss of eversion.
- 12- Tendo-achillis.
- 13- Unlocking, flexion.
- 14- Medial cuneiform bone – base of 1st metatarsal bone.
- 15- Tibialis anterior – peroneus tertius.

CROSS MATCHING

Select appropriate statement of column (B) which fits with knowledge of column (A):

1-	Column (A)	Column (B)
	a- Biceps femoris, insertion. () b- Gluteus maximus insertion. () c- Semimembranosus overlies. () d- Psoas major action. ()	1- A flexor of hip. 2- Inserted into head of fibula. 3- Iliotibial tract. 4- A bursa.
2-	Column (A) Bony origin	Column (B) Structure attached
	a- Anterior superior iliac spine. () b- Anterior inferior iliac spine. () c- Ischial tuberosity. () d- Iliac tubercle. ()	1- Origin of tensor fascia lata. 2- Origin hamstring muscles. 3- Origin of iliofemoral ligament. 4- Origin of sartorius muscle.
3-	Column (A) Bony insertion	Column (B) Muscle name
	a- Fibula. () b- Tibia. () c- Greater trochanter. () d- Base of fifth metatarsal bone. ()	1- Semitendinosus. 2- Biceps femoris. 3- Peroneus brevis. 4- Piriformis.

4-	Column (A) Muscle	Column (B) Nerve supply
	a- Biceps femoris muscle. () b- Gracilis muscle. () c- Sartorius muscle. () d- Semitendinosus muscle. ()	1- Common peroneal nerve. 2- Obturator nerve. 3- Femoral nerve. 4- Sciatic nerve.

Match the origin of the following nerves:

5-	(A)	(B)
	1- Saphenous nerve 2- Sural nerve 3- Lateral cutaneous nerve of thigh 4- Tibial nerve 5- Inferior gluteal nerve	a- Sacral plexus b- Femoral nerve c- Lumbar plexus d- Sciatic nerve e- Tibial nerve

Match the nerve supply of the following muscles:

6-	(A)	(B)
	1- Piriformis 2- Semitendinosus 3- Gastrocnemus 4- Tibialis anterior 5- Tibialis posterior	a- Sciatic nerve b- Sacral plexus c- Posterior tibial nerve d- Tibial nerve e- Deep peroneal nerve

Match column A with B:

7-	(A)	(B)
	1- Psoas major 2- Quadriceps femoris 3- Adductor magnus 4- Semimembranosus 5- Gastrocnemus	a- Tibial nerve b- Sciatic nerve c- Lumbar plexus d- Femoral nerve e- Sciatic & obturator nerves

Match the origin of the following nerves:

8-	(A)	(B)
	1- Sural communicating nerve 2- Femoral nerve 3- Nerve to obturator internus 4- Common peroneal nerve 5- Medial plantar nerve	a- Posterior tibial nerve b- Sciatic nerve c- Sacral plexus d- Lumbar plexus e- Lateral popliteal nerve

1-	a- 2	b- 3	c- 4	d- 1	
2-	a- 4	b- 3	c- 2	d- 1	
3-	a- 2	b- 1	c- 4	d- 3	
4-	a- 4	b- 2	c- 3	d- 4	
5-	1- b	2- e	3- c	4- d	5- a
6-	1- b	2- a	3- d	4- e	5- c
7-	1- c	2- d	3- e	4- b	5- a
8-	1- e	2- d	3- c	4- b	5- a

PROBLEM SOLVING

-1-

A 55 years old man was accidentally kicked with a skate on the lateral surface of his right leg just inferior to the knee. Later he experienced numbness and tingling on the lateral surface of his leg and on the dorsum of his foot and unable to dorsiflex his right foot or his toes.

1. What nerve appears to have been injured?
2. What is the relationship of this nerve to the neck of the fibula?
3. Why there is loss of sensation in the leg?
4. Name the deformity resulting from the nerve injury?
5. What are the terminal branches of the current nerve?

1. Common peroneal nerve.
2. It lies on the lateral aspect of the neck of the fibula.
3. Due to injury of its cutaneous branches.
4. Foot drop and inversion (Talipes equinovarus).
5. Superficial and deep peroneal nerves.

-2-

An old man trying to cross the road stumbled and fell and was hit by a fast crossing car. He was transferred to hospital where X-ray declared fracture of the neck of left fibula. On examining the patients lower limb he was enable to perform dorsiflexion and eversion of the left foot.

1. What is the most probable injured nerve?
2. Why the patient cannot perform dorsiflexion & eversion of left foot?
3. Name the cutaneous branches of the injured nerve.
4. Name the deformity resulting.
5. Give the origin of the injured nerve?

1. Common peroneal nerve.
2. Due to paralysis of muscles supplied by superficial & deep peroneal nerves.
3. Lateral cutaneous nerve of the calf and sural communicating nerve.
4. Foot drop and inversion (Talipes equinovarus).
5. One of the two terminal branches of sciatic nerve.

-3-

A patient was admitted to the hospital with a penetrating wound at back of thigh. Examination revealed a completely cut sciatic nerve.

1. Name the root value of the affected nerve & its origin.
2. Name 2 muscles deep to it in the gluteal region.
3. Where is the area in foot where the sensation is preserved? Why?
4. Mention 2 muscles paralyzed in the thigh.

1. L4, 5 and S1, 2, 3 arises from the sacral plexus.
2. Gemelli and Quadratus femoris.
3. Medial side of the foot till the metatarsophalangeal joint of big toe being supplied by saphenous branch of the femoral nerve.
4. Semimembranosus and biceps femoris.

-4-

A 23-year-old female is seen on the postpartum floor the day after delivering a 9-pound baby boy. She is concerned about her right foot which has become numb and weak since delivering the baby. Walking has been difficult for her because her right foot tends to drop and invert. When asked about her labor course, she reports that she had an epidural with satisfactory pain relief but a difficult

and prolonged pushing stage of labor (3 hours) in stirrups. She denies any back pain or problems with the other leg. On examination, she has decreased sensation on the dorsum of right foot and inability to dorsiflex the right foot, resulting in a foot drop. Sensation on lateral and medial sides of dorsum of the foot is intact.

1. What is the most likely diagnosis?
2. What factors likely led to this condition?
3. What is the cause of intact sensation on medial side of the foot?
4. What is the cause of intact sensation on the lateral side of the foot?
5. What is the name of this deformity?

1. Most likely diagnosis: Common peroneal nerve injury (compression).
2. Factors leading to injury: Prolonged compression of common peroneal nerve by stirrups and flexion at the knee.
3. Supplied by the saphenous nerve.
4. Supplied by sural nerve.
5. Talipes equinovarus.

-5-

A 22-year-old football player was brought to the emergency room complaining of intense pain in his left calf and ankle after trauma during a match. The left calf was tender and indurated, with an irregular mass noted in the back of the midcalf area.

1. What is the most likely diagnosis?
2. What type of excessive abnormal ankle movement would be present?
3. Is this injury rapidly heal?
4. What is the insertion of this tendon?
5. What is the thin long tendon that lies medial to Achilles tendon?

1. Most likely diagnosis: Achilles tendon rupture.
2. Likely abnormal ankle movement present: Dorsiflexion.
3. No because of poor blood supply of the Achilles tendon.
4. The posterior surface of the calcaneus at its midlevel.
5. Plantaris.

Lateral compartment of leg, foot & Joints

M.C.Q'S

- 1- The following muscle lies in the lateral compartment of the leg:**
 - a- Peroneus tertius.
 - b- Peroneus brevis.
 - c- Tibialis anterior.
 - d- Extensor hallucis longus.
- 2- The muscles supplied by the superficial peroneal nerve; mark the correct answer:**
 - a- Evert the foot.
 - b- Invert the foot.
 - c- Passes in front of the lateral malleolus.
 - d- Dorsiflex the foot.
- 3- Following muscle everts the foot:**
 - a- Tibialis anterior.
 - b- Tibialis posterior.
 - c- Extensor hallucis longus.
 - d- Peroneus brevis.
- 4- Weak inversion of foot result from damage to which muscle:**
 - a- Flexor hallucis brevis.
 - b- Flexor digitorum brevis.
 - c- Tibialis anterior.
 - d- Peroneus longus.
- 5- Superficial peroneal nerve supplies one of following muscles:**
 - a- Peroneus tertius.
 - b- Peroneus longus.
 - c- Tibialis anterior.
 - d- Flexor digitorum longus.
- 6- The following statement regarding the clinical findings of superficial peroneal nerve injury is correct:**
 - a- There is loss of planter flexion of the ankle.
 - b- There is loss of sensations of the sole of the foot.
 - c- Muscles of lateral compartment of the leg are paralyzed.
 - d- Muscles of the sole of the foot are paralyzed.
- 7- Regarding attachment of muscles of lower limb, choose correct answer:**
 - a- The gluteus maximus is inserted to the greater trochanter of femur.
 - b- The adductor longus is inserted to the linea aspra of femur.
 - c- The tibialis anterior arises from the medial surface of the tibia.
 - d- The peroneus longus is inserted to the base of 5th metatarsal bone.

- 8- Concerning nerves of the lower limb, which ONE of the following statements is true?**
- a- Sciatic nerve supplies gluteus maximus muscle.
 - b- Genito-femoral nerve supplies quadriceps femoris.
 - c- Obturator nerve supplies adductors.
 - d- Deep peroneal nerve supplies peroneal longus and brevis.
- 9- Concerning the muscles of the first layer of the sole, choose the correct statement:**
- a- Abductor hallucis lies on the lateral side.
 - b- Flexor digitorum brevis lies on the medial side.
 - c- Flexor digitorum accessorius lies in this layer.
 - d- Abductor digiti minimi lies on the lateral side.
- 10- Concerning the third layer of the sole, choose the correct statement:**
- a- Flexor hallucis brevis lies on the lateral side.
 - b- Lumbrical muscles lie in this layer.
 - c- Flexor digiti minimi lies on the medial side.
 - d- Adductor hallucis has two heads.
- 11- Regarding the medial planter nerve, select the correct answer:**
- a- It arises from the anterior tibial nerve deep to the flexor retinaculum.
 - b- It supplies adductor hallucis muscle.
 - c- It supplies flexor digitorum brevis muscle.
 - d- It supplies medial two planter interossei muscles.
- 12- Regarding muscle actions in lower limb, the following is correct statement:**
- a- Quadriceps femoris flexes the knee joint.
 - b- Abduction and adduction of the toes by the interossei take place from the middle of the second toe.
 - c- Muscles inserted in tendo calcaneus are dorsi flexors.
 - d- Gluteus maximus is a powerful flexor of hip joint.
- 13- The neck of the femur:**
- a- Wholly intracapsular.
 - b- Partially intracapsular.
 - c- Makes an angle of about 160 degrees with the shaft.
 - d- Perforated by few nutrient foramina.

14- Regarding the femur; indicate the correct answer:

- a- Its head can be palpated medial to the pulsations of femoral artery.
- b- Decrease in angle between its neck and shaft limits its abduction.
- c- Subcapital fracture of its neck is common in old age.
- d- Fracture of its lower third often interferes with blood flow to leg.

15- Regarding the fracture of the neck of the femur, select true statement:

- a- Is common in young age.
- b- Produces necrosis of the greater trochanter.
- c- Is easy to heal in old age.
- d- The limb rotates laterally after this fracture.

16- The strongest ligament of the hip joint is:

- a- Ligamentum teres.
- b- Iliofemoral ligament
- c- Pubofemoral ligament.
- d- The capsular ligament.

17- Trochanteric line gives attachment to; mark suitable answer:

- a- Rectus femoris.
- b- Vastus.
- c- Iliotibial tract.
- d- Iliofemoral ligament.

18- Hyperextension of the hip joint is limited by:

- a- Iliofemoral ligament.
- b- Ischiofemoral ligament
- c- Pubofemoral ligament.
- d- Ligament of the head of the femur.

19- Regarding the hip joint, choose the correct answer:

- a- It is a synovial joint of condyloid variety.
- b- Its extension is performed by gluteus minimus muscle.
- c- Its main flexor is the quadriceps femoris muscle.
- d- The iliofemoral ligament limits its hyperextension.

20- Regarding the adductor tubercle of the femur, choose the correct answer:

- a- Used as a point to draw the surface anatomy of the popliteal artery.
- b- Gives attachment to the medial ligament of the knee joint.
- c- Gives attachment to the adductor brevis muscle.
- d- At the lower end of medial supracondylar line.

21- The lower epiphysial plate of the femur:

- a- Is the rapidly growing end of the femur.
- b- Is closed earlier than the upper epiphysial plate.
- c- Passes through the linea aspera of the femur.
- d- The nutrient foramen is directed to it.

22- Regarding the knee joint:

- a- The tendon of popliteus muscle is intracapsular, intrasynovial.
- b- The popliteus muscle lock the knee joint at the beginning of flexion.
- c- Lateral meniscus is more liable to injury than the medial meniscus.
- d- Locking of the joint occurs at the end of extension.

23- Regarding the knee joint, the following is true:

- a- Its lateral meniscus is damaged more frequently than the medial.
- b- Its lateral meniscus is attached to the lateral collateral ligament.
- c- Rupture of its anterior cruciate ligament results in excessive backward movement of tibia on femur.
- d- Stability of joint depends largely on tone of quadriceps femoris muscle.

24- The following is inside the knee joint:

- a- Tendon of biceps femoris muscle.
- b- Cruciate ligaments.
- c- Semimembranosus bursa.
- d- Collateral ligaments.

25- Tibial collateral ligament knee extends from med. epicondyle femur to:

- a- Lateral condyle of the tibia.
- b- Medial condyle of the tibia.
- c- Articular capsule.
- d- Neck of the fibula.

26- Regarding ligaments of knee joint, mark the correct statement:

- a- The ligamentum patellae replace capsule in its anterior & lower part.
- b- Lateral collateral ligament is attached to lateral meniscus.
- c- Medial collateral ligament is not attached to the medial meniscus.
- d- In rupture of anterior cruciate ligament, there is excessive backward movement of tibia on femur.

27- In a "locked knee" which of following ligaments become taut:

- a- Fibular collateral.
- b- Tibial collateral.
- c- Anterior cruciate.
- d- Posterior cruciate.

28- The common type of football injuries in the knee joint is rupture of, choose the correct answer:

- a- Fibular collateral ligament.
- b- Posterior cruciate ligament.
- c- Plantar aponeurosis.
- d- Medial meniscus.

29- Regarding the patella, choose the correct answer:

- a- It gives attachment to the quadriceps tendon to its upper part and ligamentum patellae to the lower part.
- b- Its displacement in the medial direction is prevented by the large medial condyle of the femur.
- c- It is separated from the skin by a fascia which is liable to become inflamed, producing visible swelling.
- d- It receives lower horizontal fibers of the adductor longus on its medial side forming patellar retinaculum.

30- Regarding lower limb bones, following statements are correct:

- a- Patella is a smallest sesamoid bone.
- b- Posterior border of tibia is subcutaneous through out its entire length.
- c- Lower end of the femur possesses two condyles.
- d- Middle of third toe is taken as the axis of abduction and adduction.

31- Bone which fracture in over doing the movement of eversion is:

- a- The lateral malleolus.
- b- The calcaneum.
- c- The medial malleolus.
- d- The inferior margin of lower end of tibia.

32- If a surgeon wants to take bone chips for grafting, the most suitable place in the leg and foot is:

- a- Medial malleolus.
- b- Anteromedial surface of the tibia.
- c- The first metatarsal bone.
- d- The fibula.

33- The primary weight support function at the ankle joint is provided by which of the following bones:

- a- Fibula and tibia.
- b- Tibia and calcaneus.
- c- Calcaneus and navicular.
- d- Tibia and talus.

34- Which statement is true regarding the ankle joint:

- a- It is strengthened laterally by the deltoid ligament.
- b- It is a condylar joint.
- c- It is formed by the articulation of the talus and the distal ends of the tibia and fibula.
- d- It is most stable in the fully planter-flexed position.

35- Movements of flexion and extension of the foot occur in the following joint:

- a- Subtalar joint.
- b- Talocalcaneonavicular joint.
- c- Mid tarsal joint.
- d- Ankle joint.

36- Regarding the ankle joint, the following is correct:

- a- Sprains of the ankle are usually caused by excessive inversion of the foot.
- b- The medial or deltoid ligament is stronger than the lateral one.
- c- Its fracture dislocations is caused by forced external rotation and over eversion of the foot.
- d- Movements of inversion and eversion take place at the ankle joint.

37- The deltoid ligament; mark the correct answer:

- a- Lies on the lateral aspect of the ankle joint.
- b- Attached inferiorly to the spring ligament.
- c- Is rather a weak ligament.
- d- Is quadrilateral in shape.

38- Eversion and inversion of the foot occurs at; mark the suitable answer:

- a- Ankle joint.
- b- Subtalar joints.
- c- Inferior tibiofibular joint.
- d- First tarsometatarsal joint.

39- The movements of inversion and eversion take place at:

- a- First tarsometatarsal joint.
- b- Ankle joint.
- c- Talo-calcaneo-navicular joint.
- d- Inferior tibio-fibular joint.

- 40- Inversion of the foot is done by; mark the correct answer:**
- a- Tibialis anterior and tibialis posterior muscles.
 - b- Extensor digitorum longus.
 - c- Peroneus longus.
 - d- Peroneus tertius.
- 41- The cuboid bone; mark the most appropriate statement:**
- a- Articulates with the talus laterally.
 - b- Gives attachment to the spring ligament.
 - c- Shares in the lateral longitudinal arch.
 - d- Cylindrical in shape.
- 42- Regarding the spring ligament, choose the correct answer:**
- a- Quadrangular in shape.
 - b- Gives attachment to the deltoid ligament.
 - c- An important factor in maintaining the transverse arch of the foot.
 - d- Is called dorsal calcaneonavicular ligament.
- 43- Sesamoid bones are found in the sole in the following layer:**
- a- Second layer.
 - b- Third layer.
 - c- Fourth layer.
 - d- First layer.
- 44- The following is important factor in maintaining foot arches:**
- a- Lateral longitudinal arch is supported by tibialis posterior.
 - b- Abductor digiti minimi supports the medial longitudinal arch.
 - c- Tendon of peroneus longus supports the transverse arch.
 - d- Tendon of flexor hallucis longus supports the lateral longitudinal arch.
- 45- Regarding foot arches, the following is correct statement:**
- a- The head of talus is the "keystone" in the center of the lateral longitudinal arch.
 - b- The short & long plantar ligaments support medial longitudinal arch.
 - c- Flat foot is produced by the depression of the arches.
 - d- The cuboid is the center of the medial longitudinal arch.
- 46- Following are important factors in maintaining foot arches:**
- a- The medial longitudinal arch is supported by long plantar ligament.
 - b- The lateral longitudinal arch is supported by tibialis posterior.
 - c- Tendon of tibialis anterior suspends the lateral longitudinal arch.
 - d- Tendon of peroneus longus acts as a tie beam for transverse arch.

1- B	24- B
2- A	25- B
3- D	26- A
4- D	27- C
5- B	28- D
6- C	29- A
7- B	30- C
8- C	31- A
9- D	32- D
10- D	33- D
11- C	34- C
12- B	35- D
13- B	36- A, B & C
14- D	37- B
15- D	38- B
16- B	39- C
17- D	40- A
18- A	41- C
19- D	42- B
20- D	43- B
21- A	44- C
22- D	45- C
23- D	46- D

FILL IN THE BLANKS

- 1- Peroneus longus muscle is inserted into and
 - 2- muscle passes in a groove below the cuboid bone while muscle passes in a grooves on the back of the media malleolus.
 - 3- The tendon of peroneus longus muscle lies in the layer of the foot.
 - 4- The muscle is inserted into the tuberosity of the base of fifth metatarsal bone.
 - 5- The tendons of and pass behind then below the lateral malleolus.
 - 6- Musculocutaneous (superficial peroneal) nerve supplies 2 muscles and
 - 7- The hip bone articulates with and bones.
 - 8- The anterior superior iliac spine gives attachment to:
A-
B-
 - 9- The anterior inferior iliac spine gives attachment to:
A-
B-
 - 10- Patellar ligament extends between and
 - 11- The posterior oblique ligament of the knee is an extension of the tendon.
 - 12- The medial collateral ligament of the knee joint is attached superiorly to and inferiorly to
 - 13- Ligaments that prevent hyperextension and hyperflexion of the knee joint are and
 - 14- Fibula lies on the side of the leg and its lower end is called
 - 15- Dorsiflexion of the foot occurs at the joint while inversion occurs at joint.
 - 16- Inversion and eversion of the foot occurs at the joints.
-

- 1- Medial cuneiform bone – base of the 1st metatarsal bone.
- 2- Peroneus longus – tibialis posterior.
- 3- Fourth.
- 4- Peroneus brevis.
- 5- Peroneus longus – peroneus brevis.
- 6- Peroneus longus and brevis muscle.
- 7- Femur and sacrum.
- 8- Sartorius, inguinal ligament.
- 9- Rectus femoris (straight head), iliofemoral ligament.
- 10- Patella and tibial tuberosity.
- 11- Semimembranosus.
- 12- Medial epicondyle of the femur, medial condyle of tibia.
- 13- Anterior and posterior cruciate ligaments.
- 14- Lateral, lateral malleolus.
- 15- Ankle joint – subtalar joint.
- 16- Subtalar joints.

CROSS MATCHING

Select appropriate statement of column (B) which fits with knowledge of column (A):

1-	Column (A)	Column (B)
	a- Soleus muscle action. () b- Peroneus longus site in sole of foot. () c- Adductor hallucis site in sole of foot.() d- Tibialis anterior action. ()	1- Extensor of ankle joint. 2- Flexor of ankle joint. 3- Lies in fourth layer of foot. 4- Lies in third layer of foot.

2-	Column (A) Bone related	Column (B) Structure related or attached
	a- Cuboid. () b- Talus. () c- Sustentaculum tali. () d- Navicular. ()	1- Groove for peroneus longus tendon. 2- No muscles are attached to it. 3- Gives attachment to deltoid ligament. 4- Gives attachment to tibialis posterior.

3-	Column (A) Type of joint	Column (B) Name of joint
	a- Syndesmosis. () b- Hinge. () c- Plane. () d- Ball and socket. ()	1- Ankle joint. 2- First tarsometatarsal joint. 3- Inferior tibiofibular joint. 4- Hip joint.

Match the nerve supply of the following muscles:

4-	1- Sartorius 2- Gracilis 3- Peroneus longus 4- Flexor hallucis longus 5- Gluteus medius	a- Superior gluteal nerve b- Femoral nerve c- Posterior tibial nerve d- Superficial peroneal nerve e- Obturator nerve
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Match the bones with muscle attached:

5-	1- Calcaneus 2- 5th metatarsal 3- Medial cuneiform 4- Navicular 5- Popliteal surface of tibia	a- Tibialis posterior b- Soleus c- Peroneus brevis d- Peroneus longus e- Popliteus
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1-	a- 2	b- 3	c- 4	d- 1	
2-	a- 1	b- 3	c- 2	d- 4	
3-	a- 3	b- 1	c- 2	d- 4	
4-	1- b	2- e	3- d	4- c	5- a
5-	1- b	2- c	3- d	4- a	5- e

PROBLEM SOLVING

-1-

A grandmother slipped on the floor. Her right lower limb was laterally rotated and noticeably shorter than her left limb. She was unable to get up or lift her limb off the floor.

1. What is the common fracture site of the femur in elderly people.
 2. Why this part of bone so fragile on elderly people.
 3. Why her injured limb was shorter than the other one.
 4. What are the complications commonly associated with these fractures? and Why?
-
1. Fracture neck femur.
 2. Due to osteoporosis of old age (bone absorption is greater than bone formation).
 3. The shortening of the lower limb results from upward pull of the muscles connecting the femur to his hip bone.
 4. Nonunion & avascular necrosis of head of femur due to its poor blood supply.

-2-

During the course of a football game, a football player developed locking of the knee joint due to violent abduction and external rotation of the leg.

1. What is the most likely structure to be injured? Why?
 2. Name two extracapsular ligaments of knee joint.
 3. Name 2 muscles producing flexion of knee joint.
 4. Name the ligament that prevent the hyperextension of knee.
 5. Name ligament that prevent sliding of femur forwards during flexion of knee.
-
1. Medial meniscus because it is fixed in position by its attachment to joint capsule.
 2. Medial and lateral collateral ligaments.
 3. Semimembranosus, semitendinosus, biceps, gracilis, Sartorius (any two).
 4. Anterior cruciate ligament.
 5. Posterior cruciate ligament.

-3-

A 37-year-old male was subjected to a car accident. His left knee struck the dashboard violently, and he felt pain in his left hip. Compared with his right leg, he was noted to have a shortened left lower limb that is adducted and medially rotated. There was a painful mass in the gluteal region.

1. What is the most likely diagnosis?
2. What is the mass in the gluteal region?
3. What are the structures involved in this injury?
4. What clinically important structures are at potential risk?
5. What are the nerves that carry pain from the hip joint?

1. Posterior hip dislocation, with or without acetabular fracture.
2. The head of the femur.
3. Structures involved: Hip joint, including the femoral head, joint capsule and ligaments, and acetabulum.
4. Structures at risk: Sciatic nerve.
5. Femoral nerve (through the branch to rectus femoris), anterior division of the obturator nerve, nerve to quadratus femoris, superior gluteal nerve and sciatic nerve.

-4-

During playing tennis, a 25-year-old woman fell down and her right leg was twisted. She could not stand on her right leg due to pain. On examination, the right knee was swollen and tender. With the patient sitting on the stretcher with her knee flexed, the lower leg seemed to have several centimeters of excess anterior mobility.

1. What is the most likely diagnosis?
 2. What is the mechanism of the injury?
 3. What are the functions of the anterior cruciate ligament?
 4. What are the intracapsular structures of the knee?
 5. What are the nerves that carry pain from the knee?
-
1. Most likely diagnosis: Anterior cruciate ligament (ACL) tear.
 2. Mechanism of injury: Excessive rotational force strains or ruptures the ligament.
 3. Prevents hyperextension of the knee and checks the movement of the lateral condyle of femur or medial condyle of the tibia before the full extension so as to allow the rotation movement of locking.
 4. Ligaments (anterior and posterior cruciate ligaments, transverse ligament) menisci (medial and lateral).
 5. Femoral nerve (through the branch to vasti), posterior division of the obturator nerve, genicular branches of medial and lateral popliteal nerves.

Book MCQ's

(lower limb)

LOWER LIMB

1- From the muscles forming the floor of femoral triangle:

- a) Vastus medialis.
- b) Rectus femoris.
- c) Sartorius.
- d) Pectineus.

2- Regarding the femoral sheath mark one correct statement:

- a) Its medial compartment is occupied by the femoral artery.
- b) Its lateral compartment is occupied by the femoral nerve.
- c) The intermediate compartment is occupied by the femoral vein.
- d) It is formed of four compartments.

3- Among contents of the adductor canal is:

- a) Profunda femoris artery.
- b) Popliteal artery.
- c) Deep external pudendal artery.
- d) Saphenous nerve.

4- The powerful extension of the thigh, required when one is standing from a sitting position, is a function of the:

- a) Gluteus maximus muscle.
- b) Psoas major muscle.
- c) Iliacus muscle.
- d) Obturator externus muscle.

5- A significant weakness in abduction of the thigh would involve the following muscle:

- a) Gluteus medius.
- b) Pectineus.
- c) Gracilis.
- d) Adductor magnus.

6- Structure which passes through the greater sciatic foramen is:

- a) Quadratus femoris.
- b) Sciatic nerve.
- c) Tendon of obturator externus.
- d) Nerve to obturator internus.

7- The following structure lies deep to the gluteus maximus:

- a) Iliofemoral ligament.
- b) Obturator.
- c) Straight head of rectus femoris muscle.
- d) Greater trochanter of femur.

8- One of following structures pass through lesser sciatic foramen:

- a) Superior gluteal artery.
- b) Sciatic nerve.
- c) Obturator internus tendon.
- d) Pudendal artery.

9- Root value of sciatic nerve is:

- a) L4,5S1
- b) L4,5 S1,2
- c) L4,5 S1,2,3
- d) L2,3,4.

10- Which muscle passes through the lesser sciatic foramen:

- a) Gluteus minimus.
- b) Obturator internus.
- c) Piriformis.
- d) Quadratus femoris.

11- The structure which contributes to the above and lateral boundary of the popliteal fossa:

- a) Semimembranous muscle.
- b) Plantaris.
- c) Biceps femoris.
- d) Medial head of the gastrocnemius muscle.

12- Regarding the hamstring muscles (posterior compartment of the thigh), mark correct statement.

- a) All are supplied by common peroneal nerve.
- b) They can extend the knee.
- c) Biceps femoris produces medial rotation of knee.
- d) Upper half of semimembranous is membranous.

13- The popliteal fossa contains the following structure:

- a) Posterior tibial artery.
- b) Femoral artery.
- c) Tibial nerve.
- d) Posterior tibial nerve.

14- The muscles supplied by the superficial peroneal nerve:

- a) Evert the foot.
- b) Invert the foot.
- c) Plantar flex the toe.
- d) Dorsiflex the foot.

15- Fracture at neck of fibula results in injury of the following nerve:

- a) Common peroneal nerve.
- b) Anterior tibial nerve.
- c) Posterior tibial nerve.
- d) Femoral nerve.

16- Lesion of the medial plantar nerve could cause:

- a) Paralysis of interosseous muscles.
- b) Loss of cutaneous sensation to the adjacent sides of the plantar surface of the 1st and 2nd toes.
- c) Paralysis of flexor digitorum accessorius muscle.
- d) Paralysis of adductor hallucis muscles.

17- Which ligament limits extension of the hip joint:

- a) Iliofemoral.
- b) Pubofemoral.
- c) Ischio femoral.
- d) Round ligament of the head.

18- In injuries of the knee, the medial meniscus is frequently torn because it is firmly attached to:

- a) Anterior cruciate ligament.
- b) Fibular collateral ligament.
- c) Tibial collateral ligament.
- d) Patellar ligament

19- The thickest and strongest ligament of the hip joint is the:

- a) Ischiofemoral.
- b) Ilio-femoral.
- c) Pubo-femoral.
- d) Transverse acetabular ligament.

20- Regarding the popliteal artery

- a) It begins at the opening in the adductor canal.
- b) It ends at the lower border of the adductor longus muscle.
- c) Its pulsation can be felt while the knee is extended.
- d) It has five genicular branches.

21- Regarding the popliteal artery, select correct statement:

- a) It lies superficial to popliteal vein.
- b) It is crossed anteriorly by the tibial nerve.
- c) It terminates by dividing into anterior and posterior tibial arteries.
- d) It ends at the upper border of popliteus muscle.

22- Regarding the popliteal artery; choose one correct statement:

- a) Ends at the level of medial epicondyle of the femur.
- b) Is deeper than the popliteal vein.
- c) Gives no articular branch in the popliteal fossa.
- d) Accompanies the lateral popliteal nerve in the popliteal fossa.

23- The most usual site for feeling pulsations of dorsalis pedis artery is:

- a) Just behind the medial malleolus.
- b) Just behind the lateral malleolus.
- c) In the 1st dorsal intermetatarsal space.
- d) In the 5th dorsal intermetatarsal space.

1-d	2-c	3-d	4-a	5-a	6-b & d	7-d	8-c	9-c	10-b
11-c	12-d	13-c	14-a	15-a	16-b	17-a	18-c	19-b	20-d
21-c	22-b	23-c							

8-Concerning the dorsalis pedis artery, choose the correct answer:

- a- It is a continuation of the anterior tibial artery.
- b- It enters sole of foot by between the two heads of second dorsal interosseous muscle.
- c- It can be palpated on the dorsum of the foot between the tendons of tibialis anterior and the extensor hallucis brevis muscles.
- d- It joins the medial plantar artery.

9-The peroneal artery is a branch of which artery:

- a- Anterior tibial artery.
- b- Popliteal artery.
- c- Posterior tibial artery.
- d- Arcuate artery.

10-In the adult, the chief arterial supply of the head of femur is from:

- a- Obturator artery.
- b- Inferior gluteal artery.
- c- Deep external pudendal artery.
- d- Branches from medial and lateral circumflex femoral arteries.

11-The plantar arterial arch is formed by the lateral plantar artery and:

- a- Medial plantar artery.
- b- Anterior tibial artery.
- c- Posterior tibial artery.
- d- Dorsalis pedis artery.

12-Regarding the arteries of the lower limb; mark the correct answer:

- a- Pulsation can be readily felt from medial planter.
- b- Femoral artery gives lateral genicular branch.
- c- Middle genicular artery is a branch of the femoral artery.
- d- Head of femur is supplied by the artery accompanying ligamentum teres of the hip joint.

13-The places where the arterial pulsation in the lower limb could be felt include the:

- a- Femoral artery just below and lateral to the pubic tubercle.
- b- Popliteal artery when the knee joint is fully flexed.
- c- Posterior tibial artery below and behind the lateral malleolus.
- d- Dorsalis pedis artery on the dorsum of the foot.

14-Regarding arteries of lower limb, choose the correct answer:

- a- Profunda femoris artery gives four genicular branches.
- b- Posterior tibial artery enters the sole behind the lateral maleolus
- c- Dorsalis pedis artery is continuation of anterior tibial artery on the dorsum of the foot.
- d- Plantar arch is mainly formed by the medial plantar artery.

15-Regarding the veins of the lower limb, the following statement is correct:

- a- The great saphenous vein passes behind the medial malleolus.
- b- The small saphenous vein passes in front the lateral malleolus.
- c- Perforating veins connect the great and small saphenous veins.
- d- Perforating veins have valves, which direct the blood flow from superficial to deep veins.

16-Regarding veins of lower limb, following statement is correct:

- a- They are devoid of valves.
- b- Small saphenous vein drains into great saphenous vein.
- c- Femoral vein lies lateral to the femoral artery at the base of the femoral triangle.
- d- Great saphenous vein pierces cribriform fascia to enter femoral vein.

17-Regarding femoral artery, mark the correct statement:

- a- It lies medial to femoral vein.
- b- Its uppermost part is enclosed within femoral sheath.
- c- It enters the thigh at mid point of inguinal ligament.
- d- It passes behind the adductor longus in the adduct canal.

18-Regarding popliteal artery:

- a- It begins at the opening in the adductor canal.
- b- It ends at the lower border of the adductor longus muscle.
- c- Its pulsation can be felt while the knee is extended.
- d- It gives five genicular branches.

19-Regarding great saphenous vein, following statements are true:

- a- It starts at lateral end of dorsal venous arch.
- b- It passes behind medial malleolus.
- c- It empties into femoral vein.
- d- It has no valves.

20-Regarding the popliteal artery, select correct statement:

- a- Lies superficial to popliteal vein.
- b- Is crossed anteriorly by the tibial nerve.
- c- Terminates by dividing into anterior and posterior tibial arteries.
- d- Ends at the upper border of popliteus muscle.

21-Regarding the veins of the lower limb, choose the correct answer:

- a- Venae comitantes accompany the popliteal and femoral arteries.
- b- The long saphenous vein ascends behind the medial malleolus.
- c- Drainage of blood from deep veins is helped by contraction of calf muscles & arterial pulsation.
- d- The perforating veins possess valves which direct the blood flow from the deep to superficial veins.

1- C	12- D
2- B	13- D
3- D	14- C
4- B	15- D
5- B	16- D
6- D	17- B
7- B	18- D
8- A	19- C
9- C	20- C
10- D	21- C
11- D	

FILL IN THE BLANKS

- 1-The femoral artery begins opposit the point and ends at the
- 2-The blood supply of the head of the femur comes from:
 - A-
 - B-
- 3-The arcuate artery is a branch of the artery.
- 4-The femoral artery enters the thigh behind the inguinal ligament at a point midway between the and
- 5-The dorsalis pedis artery lies between the tendons of and
- 6-The popliteal artery begins at and ends at
- 7-The femoral artery has 3 superficial branches:, and
- 8-The femoral artery has 2 deep branches and
- 9-The vein passes in front of the medial malleolus while the vein passes behind the lateral malleolus.
- 10-Popliteal artery begins at and ends at
- 11-Anterior tibial artery begins at and ends at
- 12-Anterior and posterior tibial recurrent arteries are branches from artery while the peroneal artery is a branch from artery.

- 1- Mid inguinal, adductor hiatus.
- 2- Trochanteric anastomosis, obturator artery.

- 3- Dorsalis pedis.
- 4- Anterior superior iliac spine and symphysis pubis.
- 5- Extensor hallucis longus and Extensor digitorum longus.
- 6- Adductor hiatus, lower border of popliteus muscle.
- 7- Superficial circumflex iliac, superficial epigastric, superficial external pudendal artery.
- 8- Deep external pudendal, profunda femoris artery.
- 9- Long saphenous vein - short saphenous vein.
- 10- Adductor hiatus in adductor magnus – lower border of popliteus.
- 11- Lower border of popliteus – in front of the ankle joint midway between the 2 malleoli.
- 12- Anterior tibial – posterior tibial.

CROSS MATCHING

	Column (A)	Column (B)
1-	a-Saphenous nerve pass through. () b-Long saphenous vein when obstructed causes. () c-Descending genicular artery descends with. () d-Superior medial genicular artery arises from. ()	1- Popliteal artery. 2- Subcutaneous varicosities. 3- Subsartorial canal. 4- Femoral artery.

2- Match the origin of the following arteries:

1- Femoral artery 2- Superficial epigastric artery 3- Dorsalis pedis artery 4- Peroneal artery 5- Medial circumflex femoral	a- Internal iliac artery b- External iliac artery c- Posterior tibial artery d- Femoral artery e- Anterior tibial artery f- Profunda femoris artery
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1	a-3	b-2	c-4	d-1	
2	1-b	2-d	3-e	4-c	5-f

PROBLEM SOLVING

A 42-year-old diabetic woman complains of soreness of the left leg. She is moderately obese and has been recovering from surgical removal of her gall bladder (cholecystectomy) performed 2 weeks ago. On examination, she has obvious swelling in the left lower leg and some tenderness of the calf that increases when the calf is gently squeezed. There is no redness of the leg, and she is afebrile (without fever).

1. What is the most likely diagnosis?
 2. What structure is likely affected?
 3. What is expected to occur in the skin of the leg?
 4. What are the perforator veins?
 5. What is the importance of the calf muscles in maintaining the venous circulation in the leg?
-
1. Deep venous thrombosis (DVT).
 2. Anterior and posterior tibial veins.
 3. Occurrence of ulcers (ulcer-bearing area) due to impaired venous drainage.
 4. They are veins that lie mainly above the ankle and connect superficial veins (long and short saphenous veins) with the deep veins.
 5. In a standing position, venous return from the lower limb depends largely on muscular activity especially contraction of the calf muscles, known as the 'calf pump'. When calf muscles contract, blood is pumped proximally in the deep veins but is normally prevented from flowing into superficial veins by the valves in the perforating veins; during relaxation blood can be aspirated from superficial into deep veins.